Chapter 1: Electric Fields and Charges

Formula Sheet + Tips & Tricks

Formula Sheet

I. Electric Charge and Quantisation

Quantisation of charge

$$Q = ne$$

n = number of electrons, $e = 1.6 \times 10^{-19}$ C

II. Coulomb's Law and Superposition

Coulomb's Law

$$F = k\frac{q_1 q_2}{r^2}$$

$k = 9 \times 10^9$ N m^{2}C^{-2}; like charges repel, unlike attract

Force due to g

$$F = k\frac{e^2}{r^2} = mg$$

used to find separation at which electrostatic force equals weight

Superposition principle

$$\vec{F}_{net} = \sum \vec{F}_i$$

vector sum of individual Coulomb forces on a charge

III. Equilibrium of Charges

Null point condition

$$k\frac{q_1 q_3}{r_1^2} = k\frac{q_2 q_3}{r_2^2}$$

net force on q_3 is zero; null point lies on the side of the smaller charge if signs are opposite, and between the charges if signs are the same

IV. Electric Field Lines and Intensity

Electric field (point charge)

$$E = k\frac{q}{r^2}$$

field points away from $+q$, towards $-q$

Force on a charge in a field

$$F = qE$$

direction along E for $+q$, opposite for $-q$

Field in vector form

$$\vec{E} = k\frac{q}{r^3}\vec{r}$$

$\vec{r}$ = position vector from source charge to field point

V. Electric Dipole

Dipole moment

$$p = q \times 2l$$

$2l$ = separation between the two charges

Axial (end-on) field

$$E_{\parallel} = \frac{2kp}{r^3}$$

valid for $r \gg l$

Equatorial (broadside-on) field

$$E_{\perp} = \frac{kp}{r^3}$$

valid for $r \gg l$

Torque on a dipole

$$\tau = pE\sin\theta$$

maximum at $\theta = 90°$: $\tau_{max} = pE$

VI. Electric Flux and Gauss's Law

Electric flux

$$\phi = \vec{E} \cdot \vec{A} = EA\cos\theta$$

θ = angle between $\vec{E}$ and the outward normal $\hat{n}$

Gauss's Law

$$\phi = \frac{q_{enc}}{\varepsilon_0}$$

flux depends only on charge enclosed, not on the size/shape of the surface

VII. Linear Charge Distribution

Field due to infinite line charge

$$E = \frac{\lambda}{2\pi\varepsilon_0 r} = \frac{2k\lambda}{r}$$

λ = linear charge density (C/m)

VIII. Sheet(s) of Charge

Field due to a charged sheet

$$E = \frac{\sigma}{2\varepsilon_0}$$

field due to one plate; σ = surface charge density

Field between oppositely charged plates

$$E = \frac{\sigma}{\varepsilon_0}$$

fields of both plates add between them and cancel outside

IX. Charged Spherical Shell

Charge from surface density

$$Q = 4\pi R^2\sigma$$

R = radius of sphere

Field of a charged conducting sphere

$$E_{in} = 0; \quad E_{surface} = k\frac{Q}{R^2}; \quad E_{out} = k\frac{Q}{r^2}$$

all charge behaves as if concentrated at the centre for points outside

Tips, Tricks & Hacks

- $k = \dfrac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ Nm^{2}C^{-2} — memorise this once, it drives every Coulomb's law and field question in the chapter.
- If a question gives "force is doubled" or "distance is halved", use ratios directly: $F \propto \dfrac{1}{r^2}$, don't recompute from scratch.
- For a charge in equilibrium under gravity + electric force (classic "bead on thread" or "charged ball" problems), always resolve forces along and perpendicular to the string/incline before equating.
- Dipole field: remember End-on = 2× Broadside-on at the same distance ($E_\parallel = 2E_\perp$ for $r \gg l$) — a very common one-line MCQ trick.
- Gauss's law $\phi = \dfrac{q_{enc}}{\varepsilon_0}$ only cares about *enclosed* charge — charges outside the surface contribute zero net flux, even though they still create field at points on the surface.
- For infinite sheet/cylinder/sphere symmetry problems, choose a Gaussian surface that matches the symmetry (pillbox for sheet, cylinder for wire, sphere for point/spherical charge) — this is 90% of the trick.
- Field inside a conductor is always zero in electrostatics; all excess charge resides on the outer surface only.
- Quick sanity check: field lines start on + charges, end on − charges, never cross, and are denser where the field is stronger.